

Mixing Expert Knowledge with LLM to Improve Dialogue in Serious Games for Anorexia Caregivers: A Feasibility Study

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Abstract. Anorexia Nervosa (AN) is a severe eating disorder requiring interventions that effectively involve the entire family. Family-Based Therapy (FBT) has shown promise, yet traditional training methods for caregivers can be time-consuming and challenging. We present a serious game leveraging large language models (LLMs) that generate realistic, scenario-based dialogues between caregivers and teens with AN, offering a safe space to practice supportive communication strategies. To enhance the credibility and therapeutic relevance of these generated dialogues, we utilize a few-shot learning approach informed by domain expert feedback, enabling the model to produce contextually accurate and empathetic exchanges. By iteratively refining the system prompt with expert-validated examples, we substantially improve dialogue authenticity without the need for extensive model retraining. This approach provides a scalable, flexible solution that can be adapted to various therapeutic scenarios, ultimately broadening the reach and efficacy of digital health interventions.

Keywords. serious game, narrative, video game, carer, anorexia nervosa

1. Introduction

Anorexia Nervosa is a serious issue in adolescents, with Family-Based Therapy (FBT) recognized as an effective approach to involve the entire family in treatment [1]. Serious games have shown promise in supporting AN treatment and prevention [2, 3], and large language models can further enrich these interventions by generating realistic dialogues [4]. However, LLMs require tailored guidance to produce credible, therapeutic content.

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2. Methods

We applied a few-shot learning approach, providing four LLMs with a small set of expert-validated dialogue examples embedded in its prompt. Initially, the models generated dialogues from scenario descriptions, and a domain expert rated their credibility. Positively evaluated dialogues were then incorporated as exemplars to guide subsequent outputs. This iterative process continued until the majority of newly generated dialogues met the quality and therapeutic relevance setup by the expert.

3. Results

Initially, 54% of the reviewed dialogues were rated as credible by the expert. After refining the prompt with positively rated examples, new dialogues achieved credibility ratings up to 85%. This indicates that expert-guided few-shot learning significantly improved the model's ability to produce contextually relevant, therapeutic dialogues.

4. Discussion

Few-shot learning efficiently adjusts LLM outputs, avoiding the complexity and constraints of full model fine-tuning. Expert guidance ensures that dialogues remain clinically relevant, though limited reviewer availability and content filters posed challenges in generating consistent content. Despite these obstacles, our approach demonstrates a promising method to align LLM outputs with therapeutic goals and family-based treatment principles.

5. Conclusion

Our feasibility study shows that expert-driven few-shot learning can substantially enhance the quality of LLM-generated dialogues for caregiver training in the context of AN. Future work will expand scenario diversity and investigate the long-term impact on treatment outcomes, providing a scalable, digitally enhanced approach to family-based interventions.

References

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